

15 Some RNA molecules, called ribozymes, can catalyse reactions in a similar way to enzymes.

Many of these ribozymes have other RNA molecules as their substrates and catalyse reactions that break specific phosphodiester bonds. Another important group of ribozymes form the functional part of ribosomes.

Which statements about ribozymes and enzymes are correct?

- 1 Binding of the substrate to the active site depends on complementary pairing of bases for ribozymes and on complementary pairing of R-groups for enzymes.
- 2 Ribozymes and enzymes can catalyse hydrolytic reactions and condensation reactions.
- 3 For molecules made up of the same number of monomers, the number of possible types of enzyme is greater than the number of possible types of ribozyme.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only